

Analysis of Capacitor and Reactor Damage Rates on LCL Filters

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Abstract— DC motor drives equipped with six-pulse diode or thyristor rectifiers exhibit a substantial harmonic current distortion within a predefined threshold. To mitigate this issue, an LCL-type harmonic filter module is employed to filter the harmonics of the input current. In order to reduce the current harmonics absorbed by the power converter, an LCL filter is employed. The LCL filter comprises reactors and capacitors connected in parallel and series, with the objective of reducing the total harmonic distortion (THD-I) to a level of approximately 8%. This filter has been designed in accordance with the requirements set out in the IEC 61000-3.4 and IEEE 519 standards. The installation of an LCL filter in the Indonesia Low Speed Tunnel (ILST) system serves the purpose of reducing the harmonic current flowing through the DC converter drive section, thereby maintaining the quality of the electric power. The aforementioned filter is utilised in the low-speed wind tunnel fan drive system. This article addresses the operational characteristics of the system, the potential causes of damage to LCL filter components, and an analysis of the factors that contribute to the deterioration of these components. These factors include load characteristics, current harmonics, voltage harmonics, temperature, humidity, and environmental factors. Furthermore, the article proposes a method for early detection of damage to reactors and capacitors, which could prevent sudden or unexpected deterioration. This method represents a novel approach in this field of study.

Keywords- DC Drive System; DC Motor; LCL Filter; ILST

I. INTRODUCTION

The Laboratory for Aerodynamic, Aeroelastic, and Aeroacoustic (LA3) of the Agency for National Innovation & Research (BRIN), situated in South Tangerang, is authorised to conduct a technology feasibility study to ascertain its suitability for application in Indonesia. One of the facilities operated by LA3 BRIN is the Indonesian Low Speed Tunnel (ILST), which is used for conducting experiments in the fields of aeronautical and non-aeronautical technology. The ILST device consists of a closed wind tunnel, a direct current (DC) electric motor, and a

DC drive system [1]. The ILST device is comprised of a closed wind tunnel, a direct current (DC) electric motor, and a DC drive system. The fan drive system in the new system employs a parallel DC drive system, wherein a master motor and a slave motor are connected in parallel to a master drive system and a slave drive system, respectively. The power source for the ILST is provided by two transformers belonging to BBTA3. The installed power source capacity for LA3 is 3 MVA, which is utilized for testing activities [2]. The ILST drive system will be evaluated through a combination of field study and literature review. Each DC motor will be supplied by a transformer and DC drive system. Given the presence of DC motor drives, it is essential to understand their impact on the power quality of the system, including the power factor (PF) value [3]. With regard to literature study and system identification, data calculation and analysis has been performed to find out the existing system condition and proposed alternative solution to analyse LCL filter condition, especially in terms of capacitance and inductance value. With this knowledge, the quality and efficiency of the LCL filter system can be improved.

II. LCL FILTER

A. Conceptual Theory

Low-pass LCL filters are specifically constructed to mitigate the harmonics of current absorbed by power converters with a rectifier input stage. (Frequency converters for motors, UPS, etc.). Primarily, these filters are constructed from a parallel-series combination of reactors and capacitors, which are tailored to reduce the total harmonic distortion (I) of rectifiers. They are specifically designed to reduce the total harmonic distortion (I) to values of approximately 8%, in order to comply with the IEC-61000-3.4 and IEEE-519 standards. The selection of an LCL filter is contingent upon the current absorbed by the converter. In the case of converters with a very low power rating, it is possible to utilise a single LCL filter to supply multiple converters, provided that they are all initiated and terminated

simultaneously. In the event that multiple converters, each with its own start and stop cycle, are supplied by a single LCL filter, the system will be rendered ineffective in its ability to filter harmonics. In such a scenario, it is recommended that an individual LCL be employed for each converter.

The resonant frequency (f_{res}) of the LCL filter with a connected capacitor is given by the following equation [4] :

$$f_{res} = \frac{1}{2\pi} \sqrt{\frac{L_i + L_g}{L_i L_g C}} \quad (1)$$

where :

L_i = the inverter side inductors

L_g = the grid side inductors

C = the filter capacitor with series damping resistance

The filter capacitance (C) is selected with the objective of limiting the overrating of the inverter or reducing the inverter power factor. In its most fundamental form, the impedance of the LCL filter can be expressed as follows [4] :

$$Z_b = \frac{V_{gLL}^2}{P_r} \quad (2)$$

In this context, " V_{gLL} " is the line-to-line grid voltage and " P_r " is the rated power, respectively. The fundamental capacitance is defined as [4]:

$$C_b = \frac{1}{\omega_g Z_b} \quad (3)$$

where :

$$\omega_g = 2\pi f_0$$

The maximum allowable filter capacitance is limited to $\lambda\%$ in order to prevent excessive power factor degradation. Typically, a maximum value of $\lambda = 5\%$ is employed for grid-connected inverters, resulting in the following configuration for the filter capacitor [4] :

$$C = \lambda C_b \quad (4)$$

where :

λ = the maximum filter capacitance

In a three-phase three-system configuration, the maximum current ripple at the inverter output is given by the following equation [5] :

$$\Delta I_{Lmax} = \frac{2.2V_{dc}}{3L_{in}f_{sw}} (1 - M)M \quad (5)$$

where :

V_{dc} = DC input voltage

L_{in} = the inverter side inductors

f_{sw} = switching frekuensi

M = the SPWM modulation index.

In order to achieve the maximum permissible level of current ripple in the inverter, which is 20%, the inverter side inductance must be calculated in accordance with the following formula [5] :

$$L_{in} = \frac{4V_{dc}}{3\Delta I_{Lmax}f_{sw}} (1 - M)M \quad (6)$$

where :

V_{dc} = DC input voltage

I_{Lmax} = the maximum current ripple

f_{sw} = switching frekuensi

M = the SPWM modulation index.

The inverter side inductor serves the primary function of mitigating the switching frequency current ripple.

B. Specifications

The type of filter utilized in the ILST fan drive system to enhance the harmonic quality of DC motors, both for the master and slave units, employs a Spanish-made LCL filter, which exhibits the specifications enumerated in Table 1 [6] :

Table 1. LCL Filter Specifications

Reactor (L_1)		Capacitor (C)		Reactor (L_2)	
I	600 A	I_N	31A	I	235 A
L	0.13 mH	Q_N	3 x 246 μF	L	0.46 mH
f	50 Hz	f	50 Hz	f	50 Hz
4 sets (3 ϕ)/unit		8 banks/unit		2 sets (3 ϕ)/unit	

The configuration of the LCL filter is illustrated in the schematic presented in Figure 1 below. The basic filter is designed with L_1, L_2, L_3 and C . In some cases reactor L_4 , is already placed in front of the converter. In such case L_2 may be unnecessary.

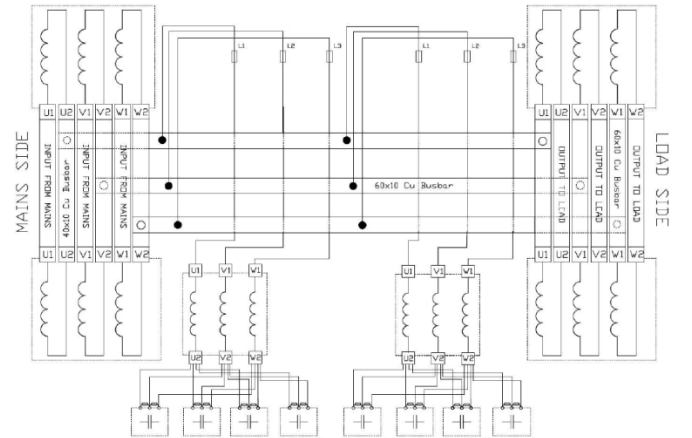


Figure 1. LCL Filter layout

The selection of LCL filters is contingent upon the current absorbed by the converter. In the event that the converters in question possess a minimal power rating, it is possible to utilise a singular LCL filter to provide power to multiple converters, provided that they are all initiated and terminated simultaneously. However, if multiple converters are initiated and terminated independently while being supplied by a single LCL filter, the system will be rendered ineffective in filtering harmonics. In such a scenario, it is advisable to employ an individual LCL filter for each converter. One disadvantage of installing the LCL filter is the voltage drop in the system that occurs due to the increase in load current. However, this is within the tolerance limit according to the standard of the installed equipment.

III. RESEARCH METHODOLOGY

Condition monitoring represents a crucial technique for assessing the condition of power electronic components, converters, and systems. It is a widely adopted approach in applications where reliability and safety are paramount, including wind turbines, electrical aircraft, and electric vehicles. Condition monitoring enables the prediction of potential failure scenarios and the implementation of preventive maintenance

strategies [7]. The condition monitoring of semiconductor devices used in power electronics has been the subject of considerable research. In addition to active semiconductor devices, capacitors are another type of component that fail more frequently than other components in power electronic systems [7]. Over the past two decades, a substantial body of scientific literature has emerged on the condition monitoring of capacitors, with a selection of the most pertinent studies discussed in this paper.

A more detailed examination of the various methods employed in the monitoring of capacitor condition, along with their respective applications, will be presented in Figure 2 [7].

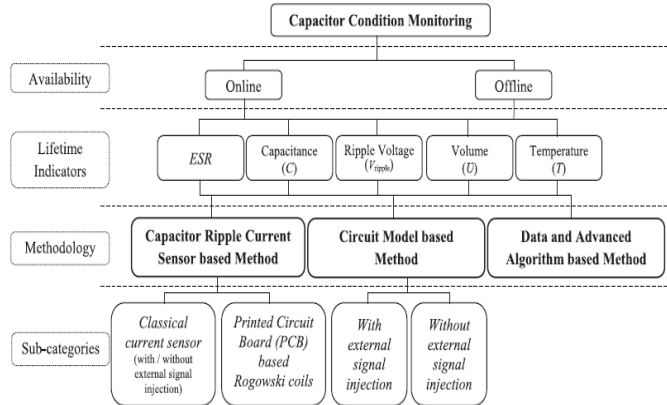


Figure 2. Capacitor Condition Monitoring

In addition to the capacitor, the inductors in the LCL filters play a pivotal role in reducing voltage distortion [8]. The LCL filter comprises two types of inductors, one situated on the grid side and the other on the converter side. These inductors serve to effectively reduce the current-THD, which is attributable to the presence of switching frequency components [8]. The converterside inductors of the LCL filter are subjected to high dv/dt , which increases the likelihood of interturn insulation degradation and eventual turn-to-turn faults. This risk is relatively higher when compared to the grid-side inductors [8].

Accordingly, Figure 3. illustrates an early fault detection and isolation technique for converter-side inductors of the LCL filter.

Techniques	Measured signals	Principle of fault detection	Capability	Disadvantages
Online techniques				
Voltage-imbalance relaying scheme [10]	Phase-to-neutral and neutral-to-ground voltages of the inductor	The summation value of phase-to-neutral and neutral-to-ground voltages under turn-to-turn fault is compared with the reference value under healthy condition	1-turn fault	Requires neutral-to-ground voltage and is not feasible for three-phase three-wire systems with supply voltage unbalance conditions
Thermography and internal temperatures [11]	Surface temperature and winding temperature	Temperature difference between the reference spot with normal temperature and a higher temperature spot	Locates hot spots due to inter-turn fault	Requires sophisticated and expensive equipment
partial discharge (PD) [11]-[13]	High frequency transient pulses in terminal voltages and currents	PD signal amplitude and number of pulses/second are used to evaluate the health condition of the transformers/inductors	Imminent insulation degradation	Expensive, equipment specific and requires complex pattern analysis
Online power factor [11], [13]	Voltage and current of each phase winding	The phase difference between terminal voltage and terminal current of each phase winding	1-turn and 2-turns faults	Only for no-load operation and requires 6-sensors
Negative-sequence currents [14]		Negative sequence currents under fault condition is compared with the reference currents under healthy operating condition	Detects inter-turn faults	Requires 6 sensors Detects fault across >3% of the total number of winding turns
Leakage fluxes [17]-[21]	Winding and core-leakage fluxes	Leakage fluxes under turn-fault/gradual degradation condition is compared with the flux under healthy conditions	1-turn to 5-turns fault and $\geq 3\%$ of the total inductance degradation detection	Dependency on the supply voltage and load conditions Unable to detect inter-turn fault at the mid-point of the winding Less sensitive to low-severity inter-turn faults
Offline techniques				
SFI ¹ and FRA ² [11], [15], [16]	Sweep frequency response of the phase windings	Frequency response of the winding under healthy and faulty conditions are compared to detect the winding anomaly	Imminent insulation degradation/inter-turn fault	Offline, requires sophisticated and expensive instruments

¹Sweep Frequency Impedance, ²Frequency Response Analysis

Figure 3. A review of the techniques employed for the detection of faults in inductors.

Following an examination of the methodologies illustrated in Figures 2 and 3, the author constructed a research flowchart, as depicted in Figure 4 below.

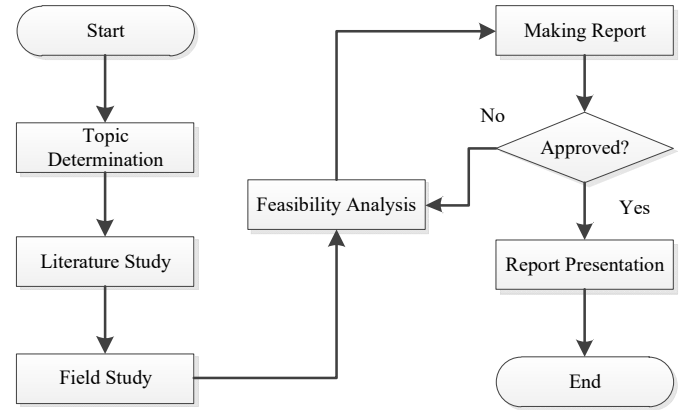


Figure 4. Research flowchart

IV. ANALYSIS OF DAMAGE RATES

The following is a review of research that affects the performance level of the LCL filter, which is designed to condition the capacitors and reactors contained therein.

A. Loading

This section delineates the current flow through the DC motor when utilizing the LCL filter, as the magnitude of the current affects the effectiveness of the LCL filter in insulating the system. Figure 5 illustrates the variation in motor current during both operational and idle states.

As indicated by the set point established by the operator, the load on the DC motor current increases. The operational range of the current is 0 amperes to a maximum value of 563.83 amperes. Based on the load characteristics, it can be concluded that the current rises with the load. Therefore, the DC motor current passing through the LCL filter is within the normal range, as it remains below the maximum current capacity of the LCL filter, which is 1200 amperes.

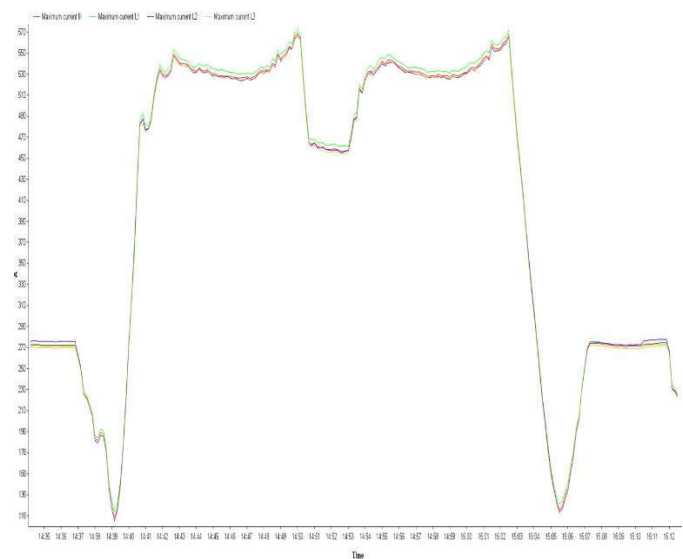


Figure 5. Characteristic DC Motor Current

B. Harmonics

The harmonic characteristics of the fan drive system can be observed and analyzed using a power quality analyzer connected to the circuit. This instrument may be employed for the purpose of recording the harmonic properties of voltage and current. Measurements were conducted on the input side of the LCL filter, encompassing both the master and slave sections of the ILST fan drive.

Table 2. THDV Limit As Per IEEE-519 Standard 1992

Power Rail Voltage At PCC	Individual Voltage Distortion (IHDv)	Total Harmonic Voltage Distortion (THDv)
≤ 69 kV	3.0	5.0
$69 \text{ kV} < V \leq 161$ kV	1.5	2.5
> 161 kV	1.0	1.5

As illustrated in Figure 6, the following depicts the voltage harmonic situation when the fan drive system is operational. As illustrated in the figure, the conditions of the 3rd, 5th, and 7th harmonics are within the normal range, as defined by the standard.

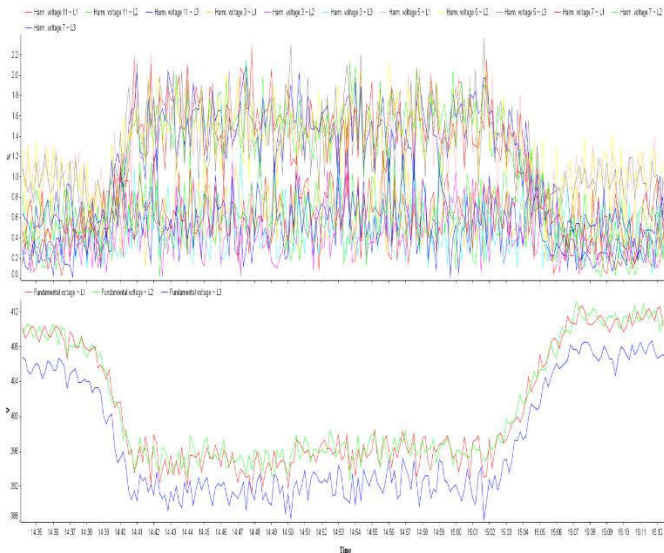


Figure 6. Characteristic of Voltage Harmonics

In accordance with the established standard in Table 2, if the voltage is less than 69 kV and the phase network fulfills the requisite voltage THD criteria, specifically a maximum individual voltage THD of 3% and a maximum total voltage THD of 5%, then the aforementioned conditions are met.

As illustrated in Figure 6, the graph demonstrates that the voltage harmonics within the LCL filter satisfy the requisite criteria and that the filter is functioning as intended, maintaining voltage harmonics in accordance with the standard.

Table 3. THDI Limit As Per IEEE-519 Standard 1992

Harmonic of Order (%)						Total Demand Distortion (TDD)
Isc/IL	<11	11-17	17-23	23-35	>35	
<20	4.0	2.0	1.5	0.6	0.3	5.0
20<50	7.0	3.5	2.5	1.0	0.5	8.0
50<100	10.0	4.5	4.0	1.5	0.7	12.0
100<1000	12.0	5.5	5.0	2.0	1.0	15.0
>1000	15.0	7.0	6.0	2.5	1.4	20.0

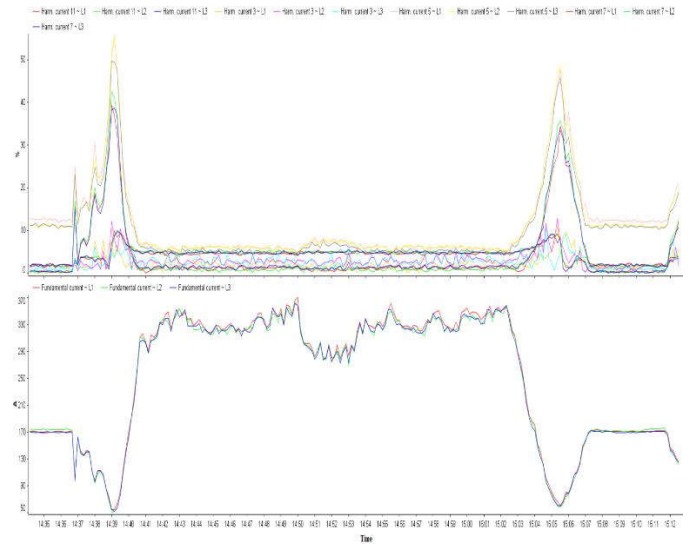


Figure 7. Characteristic of Current Harmonics

As illustrated in Figure 7, the harmonic currents flowing in each phase (L1, L2, and L3) are observed to be within the range of 5%, thereby indicating that the LCL filter is functioning optimally in reducing the harmonic currents. In accordance with the standard Table 3, if the load current flows in the range of 1000 amperes, then the 3rd, 5th, 7th, and 11th harmonics are less than 12%. Therefore, it can be stated that the harmonic condition of the current is in accordance with the standard.

Therefore, based on the analysis of the current harmonics and voltage harmonics occurring in the fan drive system, it can be concluded that the system is operating optimally, normally, and in accordance with the established standards. This indicates that the installation of the LCL filter is an appropriate solution to meet the system's requirements.

C. Temperature

The temperature rise associated with the loss of the inductor represents a significant challenge in the design of inductors for high-power-density applications. As the core loss constitutes the majority of the total inductor loss, the temperature rise of the core is a crucial thermal consideration. An effective approach is to design an inductor with a minimal core size, which can lead to acceptable losses and temperature conditions with the incorporation of a cooling system, such as a liquid cooling system [8].

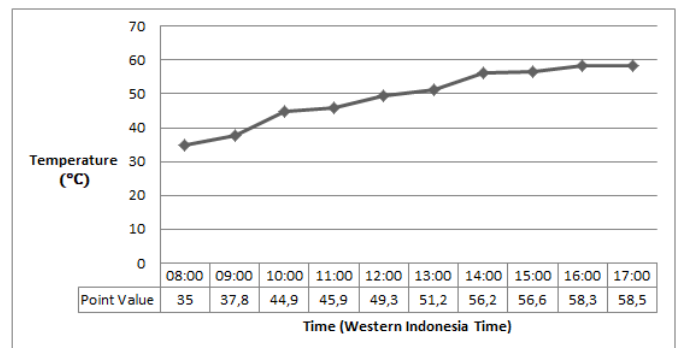


Figure 8. Temperature profile

Figure 8 illustrates that the temperature within the LCL filter reactor increases over time. Therefore, it is crucial to regularly assess the performance of the LCL filter panel at room temperature. According to the data sheet provided by the manufacturer, the optimal room temperature is 40°C. The

reactor is capable of operating at a maximum temperature of 60°C. This information elucidates the reason behind the reactor burning out, as the system's operating temperature approached the maximum temperature limit.

D. Humidity

Given the strong correlation between humidity and temperature, an increase in room humidity will result in an increase in the current flowing through the conductor, due to variations in the room temperature. It is conceivable that an increase in humidity within a given space may lead to a reduction in the insulation resistance of an apparatus, which in turn could result in an elevated leakage current. It is clear that an increase in humidity results in an overall increase in leakage current.

It is not uncommon for component aging and ambient conditions to result in significant alterations to filter capacitance. In particularly harsh environments with elevated temperatures and humidity levels, this can lead to a shift in the resonant frequency beyond the designed damping region [9].

Figure 9 illustrates the humidity levels within the LCL filter reactor. It is therefore imperative to conduct regular assessments of the LCL filter panel's functionality in ambient humidity conditions. As indicated in the manufacturer's data sheet, the optimal room humidity is 80%.

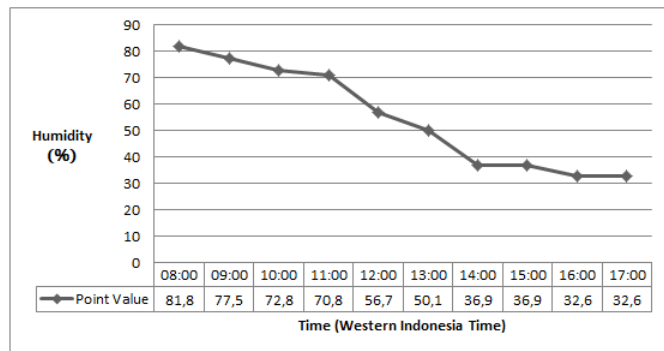


Figure 9. Humidity profile

Table 4. Measurements Of Reactor (L_1)

Phase	Unit (mH)	
	1	2
R	0,38	0,31
S	0,38	0,02
T	0,38	0,29

Table 5. Measurements Of Reactor (L_2)

Phaase	Unit (mH)			
	1	2	3	4
R	0,13	0,13	0,13	0,13
S	0,11	0,13	0,13	0,13
T	0,11	0,13	0,13	0,13

The capacitor represents a subsequent component subjected to measurement by the LCL filter. Table 6 presents the capacitor measurements obtained with the installed LCL filter.

Table 6. Measurements Of Capacitance Capacitor (C)

Phase	Unit (μF)							
	1	2	3	4	5	6	7	8
R - S	119,5	114	61,3	96	119	105,3	106,3	119,5
S - T	119,7	97	76,7	113,2	105,1	121	122	103,6
R- T	119	93,7	64,2	92,8	104,8	105	105,8	103,6

The measurement results indicate that the capacitor and reactor components of the LCL filter have likely undergone some form of damage or degradation. This equipment plays a pivotal role in maintaining the quality of harmonics, which are responsible for utilizing DC drives to remain within the predetermined range of THDV < 5% and THDI < 15%. Additionally, it is essential to ensure that the power factor remains within the range of 0.85 to prolong the lifespan of the equipment. As evidenced in Tables 4 and 5, damage to the LCL filter results in a reduction in power factor, which in turn affects the performance of the equipment. In this regard, it is essential to ascertain the condition of the equipment at an early stage. The LCL filter is comprised of two primary components: reactors and capacitors. As such, the capacitance and inductance values of these two components can be observed in real time.

E. Environment

The quality of reactor insulation is susceptible to deterioration due to fluctuations in ambient humidity. This can lead to a reduction in the mechanical strength of the insulation and an acceleration in the rate of wear. Accordingly, it is essential to monitor temperature and humidity on a continuous basis. The adhesion of contaminants, such as dirt and dust, to the reactor insulation or coils can also have a detrimental impact on the quality of the insulation, leading to a reduction in heat transfer efficiency. This is a crucial aspect that merits equal attention to that given to temperature and humidity considerations. The strength of the insulation material is susceptible to deterioration when exposed to dust, dirt, and other contaminants within the LCL filter reactor. This can lead to an increase in the failure rate of the insulation.

F. Measurement of LCL Filter

The primary objective of the fundamental research is to ascertain, elucidate, and project the data derived from the measurements. The results of the measurements of the induction reactance of the installed Reactor (L_1) and Reactor (L_2) are presented in Table 4 and Table 5.

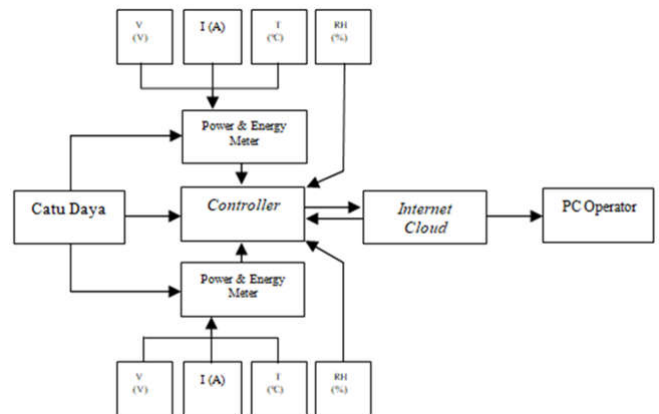


Figure 10. Module diagram of LCL Filter monitoring tool

With regard to this, the author is conducting research on the topic of Design Module Tools Monitoring Capacitance Value And Inductance Value Real Time On LCL Filter in Figure 10, which is one of the main parts of the fan drive system, ILST with the aim of maintaining system performance and improving its performance, where the ultimate goal is to be able to determine the condition of the health index or health prediction of one of the main ILST equipment by designing and building a power system equipment monitoring module, especially LCL Filter so that operators can quickly and accurately decide on emergency conditions of a piece of equipment.

CONCLUSIONS

From the foregoing discussion, it can be concluded that there are multiple factors that contribute to the deterioration of the LCL filter. These factors include the reactor inductance value, capacitance value, current harmonics, voltage harmonics, load factors, and environmental factors. Environmental factors have been identified as a significant influence on the integrity of the reactor. As a consequence of these environmental conditions, a shift in the quality of the insulation on the reactor may occur, leading to an increase in the load current on the reactor. This results in an imbalance in the harmonics of the current, which in turn gives rise to the emergence of a short circuit current in the insulation. Ultimately, this may result in the burning of the insulation winding. To prevent damage to the reactor, it is essential to develop a device that can monitor the capacitance and inductance values in real time. This tool design represents a novel approach developed by researchers.

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